GITAM
SCHOOL OF COMPUTER
SCIENCE AND ENGINEERING

GREETINGS OF THE DAY TO ALL THE PARTICIPANTS

Brief Introduction of the Instructor

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Experience	17+ Years
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Brief Introduction of the Course Structure

Name of the Course	**CRYPTOGRAPHY**
Course Code	**24CSEN3461 - CRYPTOGRAPHY**
No. of Modules	**05**
Dept. and Institute	**Dept. of AI&DS, GITAM School of CSE, GITAM, Visakhapatnam**
Regulations	**2024-25 Admitted Batch , GITAM , VSP**
Credits	**04**
Target Audience	**5th Semester CSE (CS) Students**
Prerequisites	**MATH2571 DISCRETE MATHEMATICAL STRUCTURES**

Module 1

Mathematical Foundations for Cryptography 9L + 4P

- Divisibility and the Division Algorithm
- The Euclidean Algorithm
- Modular Arithmetic
- Prime Numbers
- Fermat's and Euler's Theorems
- Testing for Primality
- The Chinese Remainder Theorem
- Discrete Logarithms
- Fields
- Finite Fields of the Form GF(p)
- Polynomial Arithmetic
- Finite Fields of the Form GF(2^n)
- Finite Field Arithmetic
- Elliptic Curve Arithmetic

Extended Euclidean Algorithm

Algorithm: Extended_Euclidean(a, b)

1. Initialize:
   r1=a, r2=b, s1=1, s2=0, t1=0, t2=1
2. While r2 ≠ 0
   q ← r1 DIV r2
   (r1,r2) ← (r2,r1 MOD r2)
   (s1,s2) ← (s2,s1−q×s2)
   (t1,t2) ← (t2,t1−q×t2)
3. Output r1 (GCD)
4. End

a(s1)+b(t1)=gcd(a,b)

A(x) + B(y)=gcd(a,b)

$$\gcd(a,b) = \begin{cases} a, & \text{if } b = 0 \\ \gcd(b, a \bmod b), & \text{otherwise.} \end{cases}$$

Example

Given : 20x+34y=gcd (20,34)

a(s1)+b(t1)=gcd(a,b)

Algorithm: Extended_Euclidean(a, b)

1. Initialize:
   r1=a, r2=b, s1=1, s2=0, t1=0, t2=1
2. While r2 ≠ 0
   q ← r1 DIV r2
   (r1,r2) ← (r2,r1 MOD r2)
   (s1,s2) ← (s2,s1−q×s2)
   (t1,t2) ← (t2,t1−q×t2)
3. Output r1 (GCD)
4. End

q	r1	r2	r	s1	s2	s	t1	t2	t
0	20	34	20	1	0	1	0	1	**0**
1	34	20	14	0	1	-1	1	0	**1**
1	20	14	6	1	-1	2	0	1	**-1**
2	14	6	2	-1	2	-5	1	-1	**3**
3	6	2	0	2	-5	17	-1	3	**-10**
X	**2**	**0**	**X**	**-5**	**17**	**X**	**3**	**-10**	**X**

s ← s1 − q × s2; t ← t1 − q × t2

Final Values are GCD(20,34) = 2 , x = s1 = -5 ; y = t1 = 3

Example with Equations

Given : 20x+34y=gcd(20,34)

1. Find the GCD

- ✓ 34 = 20 * 1 + 14
- ✓ 20 = 14 * 1 + 6
- ✓ 14 = 6 * 2 + 2
- ✓ 6 = 2 * 3 + 0

gcd(20, 34) = 2

2. Back-Substitute

- ✓ 2 = 14 - 6 * 2
- ✓ 2 = 14 - (20 - 14 * 1) * 2
- ✓ 2 = 14 * 3 - 20 * 2
- ✓ 2 = (34 - 20 * 1) * 3 - 20 * 2
- ✓ 2 = 34*3 - 20*3 – 20*2
- ✓ 2 = 34*3 – 20(3+2)
- ✓ 2 = 20 * (-5) + 34 * 3

3. Particular Solution

x = -5

y = 3

The Discrete Logarithm

A discrete logarithm is the modular arithmetic equivalent of a standard logarithm.

Given a generator g and a modulus p, it is the exponent x that solves the equation $g^x \equiv b \pmod{p}$ While discrete exponentiation g^x is computationally easy, calculating x is highly difficult and serves as the foundation for modern cryptography.

- g is the base generator.
- b is the result of the exponentiation.
- p is usually a large prime number.
- x is the unknown discrete logarithm you are trying to solve for

$$g^x \equiv b \pmod{p}$$

Example

$$3^x \equiv 13 \pmod{17}$$

$$g^x \equiv b \pmod{p}$$

Here, g=3,b=13 and prime, p =17
Check for all numbers until we match target of 13

x=1 -> 3 mod 17 - not satisfied
x=2 -> 9 mod 17 - not satisfied
x=3 -> 10 mod 17 - not satisfied
x=4 -> 13 mod 17 – satisfied

Hence, x=4

Example 1: Solve $\log_2 9 \bmod 11$.

Solution:

Here p=11, g=2, X=9

$\log_g X \equiv n \pmod p$

$X \equiv g^n \pmod p$

$9 \equiv 2^n \pmod{11}$

Try 'n' = 1, 2, 3, ...

$9 \equiv 2^6 \pmod{11}$

Feilds

A set F with operations $+$ and $\times$ is called a **Field** if:

1. F is closed under addition and multiplication.
2. Addition and multiplication are associative.
3. Addition and multiplication are commutative.
4. There exists an additive identity 0.
5. There exists a multiplicative identity 1, where $1 \neq 0$.
6. Every element has an additive inverse.
7. Every non-zero element has a multiplicative inverse.
8. Multiplication is distributive over addition.

Fields

1. Closure
For any $a, b \in F$, $a + b \in F, a \times b \in F$
Example
In $GF(7)$, $5 + 6 = 11 \equiv 4 \pmod 7$Since 4 belongs to the field, closure holds.

2. Associative Property
Addition $(a + b) + c = a + (b + c)$
Multiplication $(a \times b) \times c = a \times (b \times c)$
Example
$(2 + 3) + 4 = 2 + (3 + 4) = 9$

3. Commutative Property
Addition $a + b = b + a$
Multiplication $a \times b = b \times a$
Example

$$4 + 6 = 6 + 4$$

4. Identity Elements
Additive Identity $a + 0 = a$
Multiplicative Identity $a \times 1 = a$
Example

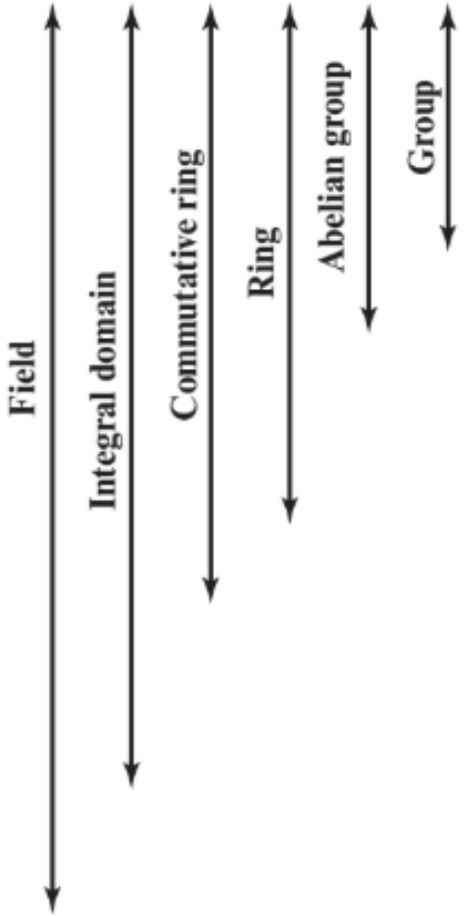
$$6 + 0 = 6$$
$$6 \times 1 = 6$$

5. Inverse Elements
Additive Inverse
Every element has an opposite.
Example
In GF(7),
Additive inverse of 5 is 2 because $5 + 2 = 7 \equiv 0$
Multiplicative Inverse
Every non-zero element has an inverse.
Example
Find inverse of 3 in GF(7).
Since $3 \times 5 = 15$and $15 \equiv 1 \pmod 7$Therefore, $3^{-1} = 5$

6. Distributive Law
$a(b + c) = ab + ac$Example

$$3(2 + 4) = 3(6) = 18$$
$$3 \times 2 + 3 \times 4 = 6 + 12 = 18$$

Fields Contd.



(A1) Closure under addition:	If a and b belong to S, then $a + b$ is also in S
(A2) Associativity of addition:	$a + (b + c) = (a + b) + c$ for all a, b, c in S
(A3) Additive identity:	There is an element 0 in R such that $a + 0 = 0 + a = a$ for all a in S
(A4) Additive inverse:	For each a in S there is an element $-a$ in S such that $a + (-a) = (-a) + a = 0$
(A5) Commutativity of addition:	$a + b = b + a$ for all a, b in S
(M1) Closure under multiplication:	If a and b belong to S, then ab is also in S
(M2) Associativity of multiplication:	$a(bc) = (ab)c$ for all a, b, c in S
(M3) Distributive laws:	$a(b + c) = ab + ac$ for all a, b, c in S $(a + b)c = ac + bc$ for all a, b, c in S
(M4) Commutativity of multiplication:	$ab = ba$ for all a, b in S
(M5) Multiplicative identity:	There is an element 1 in S such that $a1 = 1a = a$ for all a in S
(M6) No zero divisors:	If a, b in S and $ab = 0$, then either $a = 0$ or $b = 0$
(M7) Multiplicative inverse:	If a belongs to S and $a \neq 0$, there is an element a^{-1} in S such that $aa^{-1} = a^{-1}a = 1$

Figure 5.2 Properties of Groups, Rings, and Fields

Extract from Text Book

Finite Fields of the Form GF(p)

A **Galois Field (GF)** is a **finite field** containing a finite number of elements.
It is named after the French mathematician **Évariste Galois**, who developed the theory of finite fields.
A Galois field has exactly p^n elements, where:
•p is a prime number.
•n is a positive integer. (n>0)

The notation is: $\boldsymbol{GF(p^n)}$

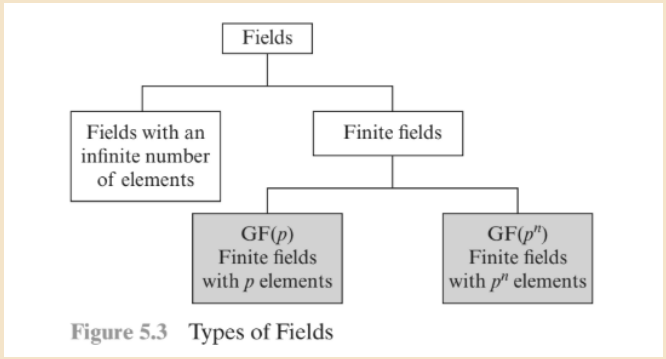


Figure 5.3 Types of Fields

SRC : TEXT BOOK

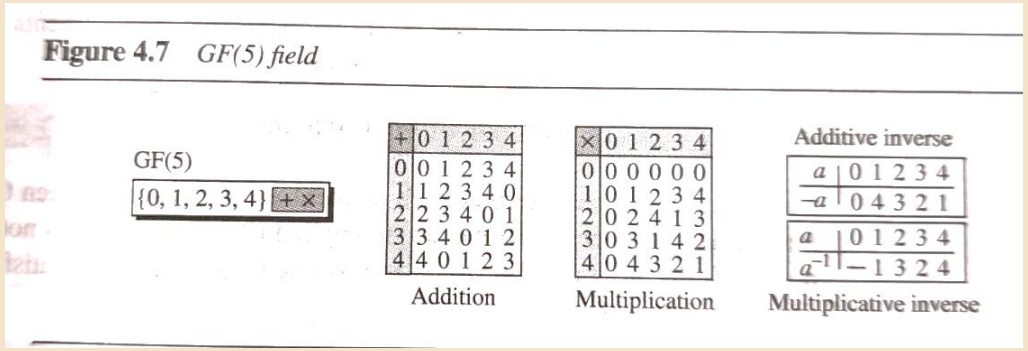


Figure 4.7 GF(5) field

Compute : $8 + 9 \pmod{11}$

Solution : $8 + 9 = 17$

Modulo 11 : $17 = 11 + 6$

Therefore, $17 \equiv 6 \pmod{11}$

Answer: 6 :: GF(11)

Compute : $4 - 9 \pmod{7}$

Solution : $4 - 9 = -5$

Add 7 until the result is non-negative:
$-5 + 7 = 2$

Therefore, $4 - 9 \equiv 2 \pmod{7}$

Answer : 2 :: GF(7)

Compute : $7 \times 9 \pmod{13}$

Solution : 63

Divide by 13 , $63 = 13 \times 4 + 11$

Hence, $63 \equiv 11 \pmod{13}$

Answer : 11 :: GF(13)

Finite Fields of the Form $GF(p^n)$

Division is performed using multiplicative inverses.

Compute : $8 \div 3 \pmod 7$

First, find inverse - $3^{-1} = 5$

Therefore, $8 \times 5 = 40$

Modulo 7

$40 \equiv 5$

Hence, $8 \div 3 \equiv 5 \pmod 7$

Answer : 5 :: GF(7)

Feature	GF(p)	GF(2^n)
Elements	Integers	Binary polynomials
Modulus	Prime number	Irreducible polynomial
Addition	Modulo p	XOR
Multiplication	Modulo p	Polynomial multiplication + reduction
Example	GF(7)	GF(2^8)
Applications	Diffie–Hellman, DSA	AES, ECC (binary fields), CRC

Polynomial Arithmetic

A polynomial in one variable $x$is written as:

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where:

- $a_i$are coefficients.
- $n$is the degree of the polynomial.

Example

$$P(x) = x^5 + x^3 + x + 1$$

- Degree = 5
- Highest power = x^5

Binary representation: 101011

Table 5.2 Arithmetic in GF(2^3)

+	000	001	010	011	100	101	110	111
	0	1	2	3	4	5	6	7
000 0	0	1	2	3	4	5	6	7
001 1	1	0	3	2	5	4	7	6
010 2	2	3	0	1	6	7	4	5
011 3	3	2	1	0	7	6	5	4
100 4	4	5	6	7	0	1	2	3
101 5	5	4	7	6	1	0	3	2
110 6	6	7	4	5	2	3	0	1
111 7	7	6	5	4	3	2	1	0

(a) Addition

×	000	001	010	011	100	101	110	111
	0	1	2	3	4	5	6	7
000 0	0	0	0	0	0	0	0	0
001 1	0	1	2	3	4	5	6	7
010 2	0	2	4	6	3	1	7	5
011 3	0	3	6	5	7	4	1	2
100 4	0	4	3	7	6	2	5	1
101 5	0	5	1	4	2	7	3	6
110 6	0	6	7	1	5	3	2	4
111 7	0	7	5	2	1	6	4	3

(b) Multiplication

w	$-w$	w^{-1}
0	0	–
1	1	1
2	2	5
3	3	6
4	4	7
5	5	2
6	6	3
7	7	4

(c) Additive and multiplicative inverses

Polynomial addition\ substraction is equivalent to the **XOR operation**

Polynomial Arithmetic Contd....

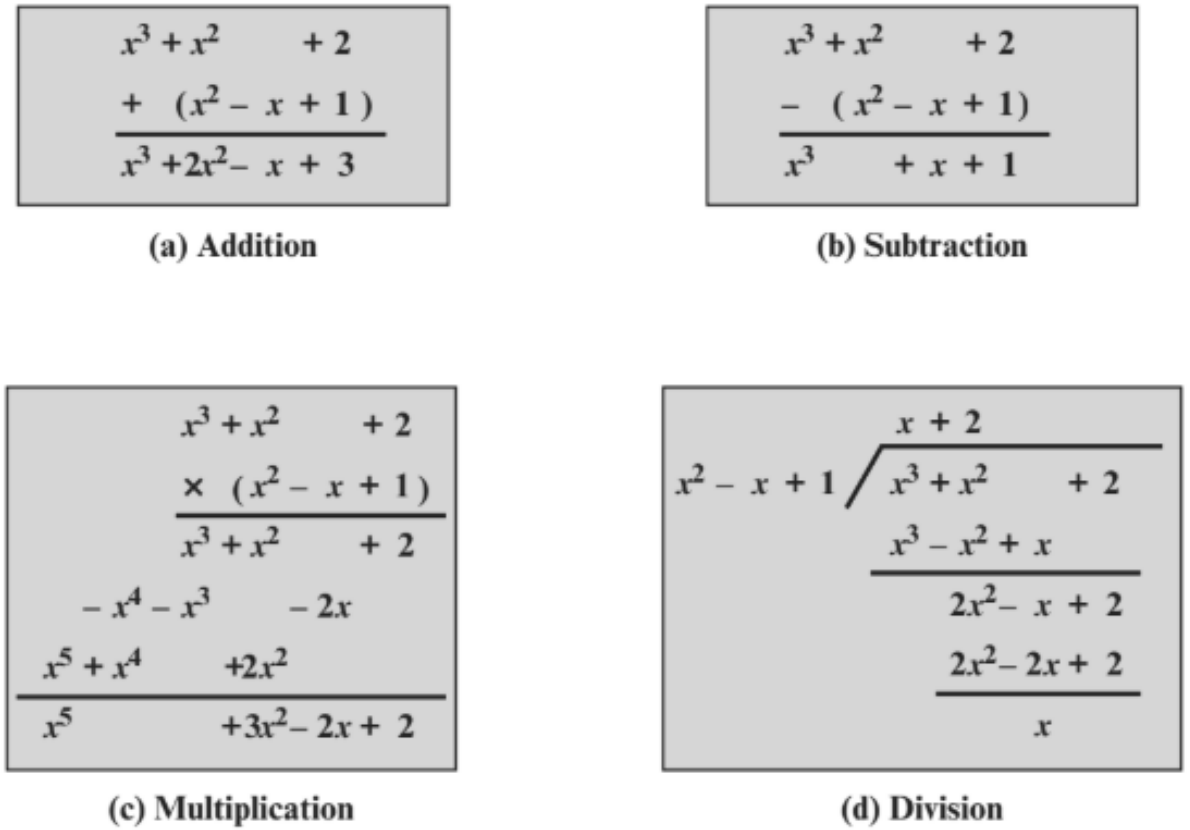


Figure 5.5 Examples of Polynomial Arithmetic

Find the GCD of

$P(x) = x^3 - 3x^2 + 2x$ &

$Q(x) = x^2 - x$.

- **Factor P(x): x(x - 2)(x - 1)**
- **Factor Q(x): x(x - 1)**
- **Common factors: x and (x - 1)**

Therefore, GCD = $x(x - 1) = x^2 - x$.

Finite Field Arithematic

Multiplication

Multiplication in polynomials is the sum of the multiplication of each term of the first polynomial with each term of the second polynomial. However, we need to remember three points. First, the coefficient multiplication is done in GF(2). Second, multiplying x^i by x^j results in x^{i+j}. Third, the multiplication may create terms with degree more than $n-1$, which means the result needs to be reduced using a modulus polynomial. We first show how to multiply two polynomials according to the above definition. Later we will see a more efficient algorithm that can be used by a computer program.

Example 4.19

Find the result of $(x^5 + x^2 + x) \otimes (x^7 + x^4 + x^3 + x^2 + x)$ in GF(2^8) with irreducible polynomial $(x^8 + x^4 + x^3 + x + 1)$. Note that we use the symbol $\otimes$ to show the multiplication of two polynomials.

Solution

We first multiply the two polynomials as we have learned in algebra. Note that in this process, a pair of terms with equal power of x are deleted. For example, $x^9 + x^9$ is totally deleted because the result is a zero polynomial, as we discussed above.

$$P_1 \otimes P_2 = x^5(x^7 + x^4 + x^3 + x^2 + x) + x^2(x^7 + x^4 + x^3 + x^2 + x) + x(x^7 + x^4 + x^3 + x^2 + x)$$
$$P_1 \otimes P_2 = x^{12} + x^9 + x^8 + x^7 + x^6 + x^9 + x^6 + x^5 + x^4 + x^3 + x^8 + x^5 + x^4 + x^3 + x^2$$
$$P_1 \otimes P_2 = (x^{12} + x^7 + x^2) \bmod (x^8 + x^4 + x^3 + x + 1) = x^5 + x^3 + x^2 + x + 1$$

To find the final result, divide the polynomial of degree 12 by the polynomial of degree 8 (the modulus) and keep only the remainder. The process is the same as we have learned in algebra, but we need to remember that subtraction is the same as addition here. Figure 4.10 shows the process of division.

Figure 4.10 *Polynomial division with coefficients in GF(2)*

$$\begin{array}{r|l} & x^4 + 1 \\ \hline x^8 + x^4 + x^3 + x + 1 & x^{12} + x^7 + x^2 \\ & x^{12} + x^8 + x^7 + x^5 + x^4 \\ \hline & x^8 + x^5 + x^4 + x^2 \\ & x^8 + x^4 + x^3 + x + 1 \\ \hline \text{Remainder} & \boxed{x^5 + x^3 + x^2 + x + 1} \end{array}$$

Table 4.4 *List of irreducible polynomials*

Degree	*Irreducible Polynomials*
1	$(x + 1)$, (x)
2	$(x^2 + x + 1)$
3	$(x^3 + x^2 + 1)$, $(x^3 + x + 1)$
4	$(x^4 + x^3 + x^2 + x + 1)$, $(x^4 + x^3 + 1)$, $(x^4 + x + 1)$
5	$(x^5 + x^2 + 1)$, $(x^5 + x^3 + x^2 + x + 1)$, $(x^5 + x^4 + x^3 + x + 1)$, $(x^5 + x^4 + x^3 + x^2 + 1)$, $(x^5 + x^4 + x^2 + x + 1)$

Src: Cryptography Textbook – Behrouz A Forouzan

Polynomial Arithematic

Multiplicative Inverse Finding the multiplicative inverse is a little more involved. The extended Euclidean algorithm must be applied to the modulus and the polynomial. The process is exactly the same as for integers.

Example 4.20

In **GF**(2^4), find the inverse of $(x^2 + 1)$ modulo $(x^4 + x + 1)$.

Solution

We use the extended Euclidean algorithm as in Table 4.5:

Table 4.5 *Euclidean algorithm for Exercise 4.20*

q	r_1	r_2	r	t_1	t_2	t
$(x^2 + 1)$	$(x^4 + x + 1)$	$(x^2 + 1)$	(x)	(0)	(1)	$(x^2 + 1)$
(x)	$(x^2 + 1)$	(x)	(1)	(1)	$(x^2 + 1)$	$(x^3 + x + 1)$
(x)	(x)	(1)	(0)	$(x^2 + 1)$	$(x^3 + x + 1)$	(0)
	(1)	(0)		$(x^3 + x + 1)$	(0)	

This means that $(x^2 + 1)^{-1}$ modulo $(x^4 + x + 1)$ is $(x^3 + x + 1)$. The answer can be easily proved by multiplying the two polynomials and finding the remainder when the result is divided by the modulus.

$$[(x^2 + 1) \otimes (x^3 + x + 1)] \bmod (x^4 + x + 1) = 1$$

Example 4.21

In GF(2^8), find the inverse of (x^5) modulo $(x^8 + x^4 + x^3 + x + 1)$.

Solution

Use the Extended Euclidean algorithm as shown in Table 4.6:

Table 4.6 *Euclidean algorithm for Example 4.21*

q	r_1	r_2	r	t_1	t_2	t
(x^3)	$(x^8 + x^4 + x^3 + x + 1)$	(x^5)	$(x^4 + x^3 + x + 1)$	(0)	(1)	(x^3)
$(x + 1)$	(x^5)	$(x^4 + x^3 + x + 1)$	$(x^3 + x^2 + 1)$	(1)	(x^3)	$(x^4 + x^3 + 1)$
(x)	$(x^4 + x^3 + x + 1)$	$(x^3 + x^2 + 1)$	(1)	(x^3)	$(x^4 + x^3 + 1)$	$(x^5 + x^4 + x^3 + x)$
$(x^3 + x^2 + 1)$	$(x^3 + x^2 + 1)$	(1)	(0)	$(x^4 + x^3 + 1)$	$(x^5 + x^4 + x^3 + x)$	(0)
	(1)	(0)		$(x^5 + x^4 + x^3 + x)$	(0)	

This means that $(x^5)^{-1}$ modulo $(x^8 + x^4 + x^3 + x + 1)$ is $(x^5 + x^4 + x^3 + x)$. The answer can be easily proved by multiplying the two polynomials and finding the remainder when the result is divided by the modulus.

$$[(x^5) \otimes (x^5 + x^4 + x^3 + x)] \bmod (x^8 + x^4 + x^3 + x + 1) = 1$$

Example:

https://drive.google.com/file/d/1qijDvDrr4tngsOTkHJc0xG9D1E7-Q2PG/view?usp=drive_link

Table 4.9 *Addition table for GF(2^3)*

⊕	000 (0)	001 (1)	010 (x)	011 (x + 1)	100 (x^2)	101 $x^2 + 1$	110 ($x^2 + x$)	111 ($x^2 + x + 1$)
000 (0)	000 (0)	001 (1)	010 (x)	011 (x + 1)	100 (x^2)	101 ($x^2 + 1$)	110 ($x^2 + x$)	111 ($x^2 + x + 1$)
001 (1)	001 (1)	000 (0)	011 (x + 1)	010 (x^2)	101 ($x^2 + 1$)	100 ($x^2 + x$)	111 ($x^2 + x + 1$)	110 ($x^2 + x$)
010 (x)	010 (x)	011 (x + 1)	000 (0)	001 (1)	110 ($x^2 + x$)	111 ($x^2 + x + 1$)	100 ($x^2 + x$)	101 ($x^2 + 1$)
011 (x + 1)	011 (x + 1)	010 (x)	001 (1)	000 (0)	111 ($x^2 + x + 1$)	110 ($x^2 + x$)	101 ($x^2 + 1$)	100 (x^2)
100 (x^2)	100 (x^2)	101 ($x^2 + 1$)	110 ($x^2 + x$)	111 ($x^2 + x + 1$)	000 (0)	001 (1)	010 (x)	011 (x + 1)
101 ($x^2 + 1$)	101 ($x^2 + 1$)	100 (x^2)	111 ($x^2 + x + 1$)	110 ($x^2 + x$)	001 (1)	000 (0)	011 (x + 1)	010 (x)
110 ($x^2 + x$)	110 ($x^2 + x$)	111 ($x^2 + x + 1$)	100 (x^2)	101 ($x^2 + 1$)	010 (x)	011 (x + 1)	000 (0)	001 (1)
111 ($x^2 + x + 1$)	111 ($x^2 + x + 1$)	110 ($x^2 + x$)	101 ($x^2 + 1$)	100 (x^2)	011 (x + 1)	010 (x)	001 (1)	000 (0)

Table 4.10 *Multiplication table for GF(2^3) with irreducible polynomial* $(x^3 + x^2 + 1)$

⊗	000 (0)	001 (1)	010 (x)	011 (x + 1)	100 (x^2)	101 ($x^2 + 1$)	110 ($x^2 + x$)	111 ($x^2 + x + 1$)
000 (0)	000 (0)	000 (0)	000 (0)	000 (0)	000 (0)	000 (0)	000 (0)	000 (0)
001 (1)	000 (0)	001 (1)	010 (x)	011 (x + 1)	100 (x^2)	101 ($x^2 + 1$)	110 ($x^2 + x$)	111 ($x^2 + x + 1$)
010 (x)	000 (0)	010 (x)	100 (x^2)	110 ($x^2 + x$)	101 ($x^2 + 1$)	111 ($x^2 + x + 1$)	001 (1)	011 (x + 1)
011 (x + 1)	000 (0)	011 (x + 1)	110 ($x^2 + x$)	101 ($x^2 + 1$)	001 (1)	010 (x)	111 ($x^2 + x + 1$)	100 (x^2)
100 (x^2)	000 (0)	100 (x^2)	101 ($x^2 + 1$)	001 (1)	111 ($x^2 + x + 1$)	011 (x + 1)	010 (x)	110 ($x^2 + x$)
101 ($x^2 + 1$)	000 (0)	101 ($x^2 + 1$)	111 ($x^2 + x + 1$)	010 (x)	011 (x + 1)	110 ($x^2 + x$)	100 (x^2)	001 (1)
110 ($x^2 + x$)	000 (0)	110 ($x^2 + x$)	001 (1)	111 ($x^2 + x + 1$)	010 (x)	100 (x^2)	011 (x + 1)	101 ($x^2 + 1$)
111 ($x^2 + x + 1$)	000 (0)	111 ($x^2 + x + 1$)	011 (x + 1)	100 (x^2)	110 ($x^2 + x$)	001 (1)	101 ($x^2 + 1$)	010 (x)

Elliptic Curve Arithmetic

Elliptic curve arithmetic uses algebraic point operations over finite fields to build fast, small-key public-key cryptography. The core operations are point addition, point doubling, and scalar multiplication. Its security relies on the hardness of the Elliptic Curve Discrete Logarithm Problem (ECDLP).

Mathematical Foundations :

- **Weierstrass Equation:** Curves are defined by equations $y^2 = x^3 + ax + b$ over finite fields.
- More details will be discussed later.

References

- Text Book – William Stallings, Behrouz A Forouzan
- Lecture Notes available in Internet & Faculty Resources
- NPTEL and other MOOC Videos
- GeekforGeeks for Coding illustration

Thank You

Best Wishes
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